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zOS – Zero OS*

Bare minimum. Verified. Real-time. An OS for the AI era.

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* zOS is a working title. Before its first release the project gets a name of its own, one that treads on nobody's toes, mainframes included.

Abstract. This paper proposes zOS, the zero operating system: an operating system reduced to what cannot be removed. zOS does not try to make AI trustworthy. It is built to stay safe although the agent is not: the agent never holds more authority than a person granted, and no motion leaves a verified loop. zOS consists of a bootloader, a formally verified microkernel, and the bare minimum needed to run AI agents safely and in real time. Nothing at its base is new: the seL4 microkernel with its mixed-criticality extensions, its system framework and drivers, capability-based security, open roots of trust and open agent protocols all exist as open-source work. What has to be built is the layer between agents and the world: a broker that turns human intent into bounded authority, and a safety path that keeps every motion under a verified loop. One rule binds the system: whatever is not needed to run agents is left out, and for every part that remains we state whether a guarantee depends on it and whether it is proven, tested or assumed.

The history of 1991 frames the case; three shifts visible in 2026 carry it, together with the physical turn of agents into robots, which makes real time a first principle rather than an option. zOS therefore starts where agents act physically: one robot, one task. We also state where the design can fail. The horizon is deliberately short: five years, to 2031.

Keywords: microkernel, seL4, formal verification, real time, mixed criticality, object capabilities, AI agents, robotics, open source

In brief

AI agents no longer just give answers; they deliver work. They read email, hold credentials, call tools and, increasingly, move arms or wheels in the physical world. Run as an ordinary program, an agent can inherit broad access to its user's files, credentials and network; sandboxes narrow that as a layer added on top. And an agent can be misled by anything it reads. In robots, the agent's software and the real-time control of the motors mostly live in separate worlds, joined by glue.

We do not trust the agent, so authority and safety move beneath it. zOS treats every agent as untrusted, whatever model drives it. An agent holds only the authority a person granted it, limited in scope and time, and revocable. It may propose any motion; a small, deterministic loop has the last word. Processor time is capability-controlled like memory, so an agent can be prevented from monopolising the processor and starving the control path. Verified isolation, explicit capabilities and hard real time form one architecture, not three layers. Every part a guarantee depends on is labelled proven, tested or assumed: proof where feasible, explicit assumptions everywhere else.

The building blocks exist. seL4, whose MCS configuration on 64-bit RISC-V now has a machine-checked proof of functional correctness; a verified real-world Ethernet driver; OpenTitan roots of trust, shipping in production Chromebooks since March 2026. The hard part is composing them, with an end-to-end assurance argument that holds across

every seam between proven and unproven parts. Known gaps include verified multicore configurations and timing bounds for modern processors. This paper starts that argument and says where it does not know yet.

Where it can fail. It does not stop prompt injection; it limits how far a deceived agent can reach. Firmware and GPUs remain trusted, not proven. And hardened Linux beside a separate motor controller may prove good enough.

What we need now. People who know firmware, verification or robots, to test this argument before the architecture hardens around it.

Prologue: Imagine it's 1991

Imagine it's 1991. We are about to release a few lines of code into a world that will be very different from today.

On 6 August 1991, a researcher at CERN posts a note about a project called WorldWideWeb to the newsgroup alt.hypertext and invites others to join (W3C 2016). Nineteen days later, on 25 August, a student at the University of Helsinki writes to comp.os.minix: *"I'm doing a (free) operating system (just a hobby, won't be big and professional like gnu) for 386(486) AT clones"* (Torvalds 1991). The first code follows in September. Neither author knows what he has started. The web will connect the planet; the hobby will end up in servers, phones and much of what lies in between.

What made that moment fertile was not brilliance alone. It was a convergence: cheap hardware nobody owned, a network that let strangers collaborate, a licence that kept the work open, and an idea small enough to start that evening.

Now imagine it's 2026. The machines that matter are no longer only desks and servers. They are agents: programs that read, decide and act on their own, and increasingly do so with arms, wheels and rotors. They all run on operating systems built for a person at a keyboard. Finding flaws in software has become cheap, because machines now do it. Proving software correct is becoming cheaper, for the same reason. Almost everything one would need to build a new foundation already exists, scattered across open-source projects, waiting to be assembled.

We are, once again, about to release a few lines of code into a world that will be very different from today. This paper asks what those lines should be. Our answer is: as few as possible.

1 The idea: zero as a design rule

Don't trust the agent. Verify the system. zOS assumes nothing about the agent's good behaviour. It may be wrong, manipulated or hostile: in the terms of distributed systems, a Byzantine component, one that may fail in any way at all (Lamport et al. 1982). Safety therefore cannot come from the agent. It has to come from the parts that decide what the agent may do and what moves, and those parts must be proven, not trusted. That decides the size of the system. The requirement was written down more than fifty years ago: the reference monitor, the mechanism that checks every access, must always be invoked, must be tamper-proof, and must be small enough to be subject to analysis and tests (Anderson 1972). Everything in zOS follows from taking the last condition seriously.

zOS stands for zero OS. The name is a design rule, not a slogan: start from nothing and add only what cannot be left out. Three things cannot be left out.

1. **A bootloader** that brings up the machine, checks what it loads, shows what it found, and hands over to the kernel or stops.
2. **A microkernel** that isolates components, moves messages between them, and schedules time. Nothing else runs at its privilege level. Firmware beneath it runs higher still and is named as trusted.
3. **The bare minimum to run agents:** a way to reach a model, a way to reach tools, a broker that hands out authority, and real-time loops for anything that touches the physical world.

Everything else, from file systems and window managers to compatibility with existing software, lives outside the core or does not exist at all. Where legacy is unavoidable, it runs as a guest in an isolated virtual machine.

One rule binds the components. For every part, zOS answers two questions. **Does a guarantee depend on it?** If so, it belongs to the trusted base, even when it runs isolated: a broker that grants the wrong authority makes the system unsafe while the kernel enforces every boundary correctly. Isolation limits what a failing part can damage; it does not make the part right. **What do we know about it?** Its behaviour is proven against a specification, tested, or assumed. The aim is a trusted base that is small and, step by step, proven. What must stay assumed, such as firmware, boot code and devices that write to memory directly, is named and listed. A design that pretends otherwise is lying (Section 6).

The second principle is time. A robot arm, a drone or a vehicle does not fail gracefully when a deadline is missed. zOS therefore treats time as a resource protected by the kernel, like memory, from the first line of code. Provable timing is a goal zOS works towards, not a property it can claim today (Section 4).

The third principle is that zOS invents nothing at its base. Section 5 shows that the base exists as open-source work. What has to be built is the layer that connects agents to authority and to motion, and the evidence that the assumptions of the parts fit together.

The fourth is focus. Minimality only means something against a task: what cannot be left out is decided by what the device must do. zOS therefore starts where agents act in the physical world, with one robot, one task and defined operating conditions (Section 7). A general operating system for agents is the horizon, not the starting point.

2 Why now

2.1 What 1991 teaches

On 29 January 1992, Andrew Tanenbaum declared Linux obsolete in the same newsgroup: it was monolithic and tied to one processor family (DiBona et al. 1999). Many in operating-systems research found his design argument the stronger one. Linux won anyway: on cheap hardware, under the GPL, by accepting contributions pragmatically, and by running the software people already had. zOS gives up that compatibility in its core and buys it back through Linux guests (Section 3). That is a deliberate trade, not a free one.

The microkernel did not lose on its own, though. Mach, the flagship of the 1980s, was slow. Jochen Liedtke rebuilt the idea from scratch and made inter-process communication up to 22 times faster than Mach for short messages (Liedtke 1993); against contemporary microkernels the gap was roughly a factor of 10 to 20 (Heiser and Elphinstone 2016). From this L4 line came seL4, the first general-purpose kernel with a machine-checked proof of functional correctness (Klein et al. 2009). Mach survives mainly inside Apple's hybrid XNU kernel (Apple n.d.).

The lesson is double. A good design is necessary but not sufficient: timing, licence, drivers and distribution decide. And the better design needs a moment in which its cost falls. History cannot prove that 2026 is such a moment; the three shifts below argue that it may be.

2.2 Finding flaws has become cheap

In the final of DARPA's AI Cyber Challenge in August 2025, autonomous systems found 54 of 63 planted vulnerabilities, patched 43 and discovered 18 real ones (DARPA 2025). In April 2026, Anthropic reported that an unreleased model had found thousands of high-severity vulnerabilities, including some in every major operating system and web browser, among them a 27-year-old flaw in OpenBSD (Anthropic 2026a). A month later the company and around 50 partners reported more than 10,000 findings rated high or critical; of the 1,752 assessed further, 62 per cent were confirmed at that severity. These are the company's own figures; the bottleneck, it reported, had moved from finding to verifying, disclosing and patching (Anthropic 2026b).

When finding flaws is cheap, every line of unproven privileged code becomes a liability. The Linux kernel source approached 40 million lines in early 2025, about 25 million of them drivers (Leemhuis 2025). An installation runs only part of that tree, but all of it with full privilege. The same tools also help harden existing systems (Section 6); for systems that must not fail, we argue for less privileged code, not only more patches.

2.3 Proving correctness is becoming cheaper

seL4's first proof covered about 8,700 lines of C and took, together with design and implementation, roughly 25 person-years, much of it new research and tools; the authors put a repeat at under ten (Klein et al. 2009; Trustworthy Systems n.d.-a). Two developments now push that cost down. Verification tools for Rust such as Verus check systems code 3 to 61 times faster than earlier tools (Lattuada et al. 2024). And language models, paired with a proof checker that rejects every wrong step, generate proofs: a neuro-symbolic method proves 77.6 per cent of the seL4 theorems in the FVEL benchmark, though only about 20 per cent of those whose proofs exceed ten lines (He et al. 2026). Short proofs are getting cheap; long ones, and the invariants behind them, are not yet.

2.4 Agents act, and increasingly in the physical world

Classical operating systems grant a process the full authority of its user. Norm Hardy described in 1988 why this fails: a program acting on behalf of one party misuses authority granted by another, a *confused deputy* (Hardy 1988). Indirect prompt injection is the same pattern: instructions hidden in data steer an agent that acts with its user's rights (Greshake et al. 2023). Heuristic defences do not hold: adaptive attacks broke 12 published defences, most of them with success rates above 90 per cent (Nasr et al. 2026). The structural answer is old: authority must be passed explicitly, as capabilities (Dennis and Van Horn 1966; Miller 2006).

At the same time, agents are acquiring bodies. Robot stacks now separate a reasoning model that plans and calls tools from an action model that drives the motors (Google 2025). Action models emit motor commands in chunks so that a robot can hold a 50 Hz control rate while the slower model thinks (Black et al. 2025). And a project for US Army unmanned ground vehicles has built proofs of concept that run ROS 2, the common robot middleware, on seL4 (Lide 2026). For such systems a missed deadline is not a slow web page. It is a dropped load or a collision.

3 Anatomy of zOS

zOS has six layers. Read from the bottom, each one exists only because the one above cannot work without it.

Hardware and root of trust. zOS needs memory management, an IOMMU for devices that write to memory directly, and a root of trust that measures the first code to run. An open option exists: OpenTitan, an open-source silicon root of trust, has shipped in production Chromebooks since March 2026 (Google 2026). Where available, CHERI adds unforgeable pointers in hardware; it is being standardised for RISC-V (Moore 2026). Firmware runs here too, and with more privilege than the kernel: on RISC-V, seL4 Microkit relies on SBI firmware such as OpenSBI in machine mode (seL4 Microkit n.d.), on Arm the equivalent is trusted firmware at the highest exception level. This layer belongs to the trusted base and is *assumed*. zOS states a hardware profile instead of assuming any hardware will do.

Bootloader. It checks one signed, static system image and starts it, or stops. That is its only rule: no menu, no configuration, no fallback, and nothing of it stays resident. It also draws the first screen, because when the check fails nothing else will run to say why. The screen shows only compiled-in text and what the loader has measured itself, and marks as assumed what it cannot check, the loader included. Boot code lies outside seL4's verified configurations (seL4 Foundation n.d.-a), so this layer, too, is trusted base and *assumed*,

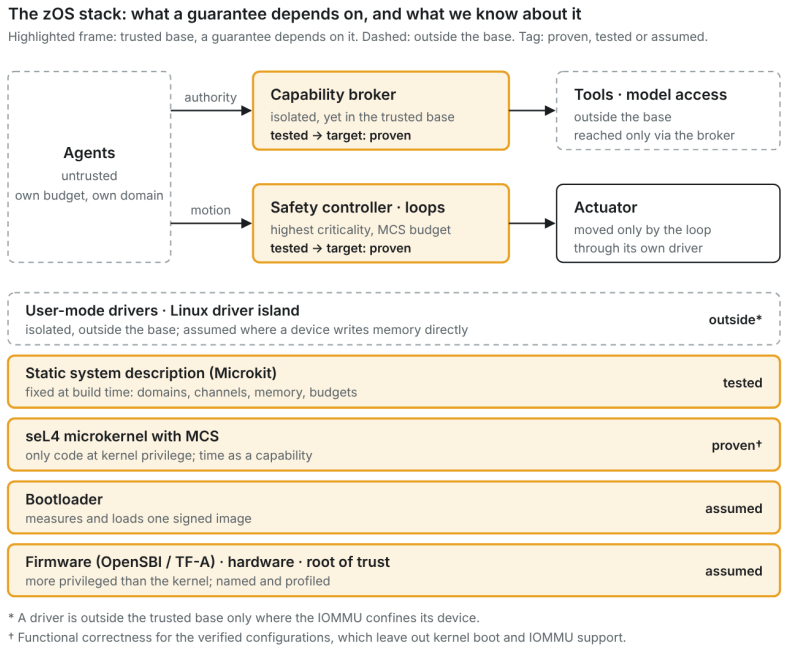


Figure 1. The zOS stack. A gold frame marks the trusted base: every part a guarantee depends on, isolated or not. The tag says what we know about it. Authority runs from the agent through the broker, motion through the safety controller; there is no third path.

and its size counts: a first prototype for RISC-V, which so far checks a hash rather than a signature, is just under 9 KB, about two thirds of it the renderer and its font data.

Microkernel. seL4 with its mixed-criticality (MCS) extensions. MCS treats time, like memory, as a resource protected by capabilities (Heiser 2026). Its C-level verification is complete for RISC-V; AArch64 is targeted for the third quarter of 2027 (seL4 Foundation n.d.-b). It is the only code at the kernel's privilege level, and its functional correctness is *proven* for the verified configurations, which leave out kernel boot and IOMMU support (seL4 Foundation n.d.-a).

Static system. The whole system is declared in one description: which isolated programs exist (protection domains), which may talk (channels) and which memory they share. A tool turns that description into the bootable image (seL4 Microkit n.d.). Nothing is loaded into the core at run time. A system whose structure is fixed at build time can be checked, scheduled and audited before it ever runs.

Drivers. Drivers run in user mode, each with access only to its own device registers (Parker 2022). Legacy drivers, above all for graphics and AI accelerators, run inside a Linux virtual machine that serves as a driver island (Heiser et al. 2025). These parts are isolated as far as memory goes and stay outside the trusted base. Wherever a device can write to memory directly and the IOMMU configuration is not proven, the driver joins the trusted base as *assumed* and is listed as such. The island sits on the agent's side: neither the authority path nor the motion path runs through it. If it fails, usefulness and confidentiality suffer, not physical safety; confidentiality does, because local inference sends the model's context through it.

The agent minimum. Four components, no more:

1. **Model access:** local inference on an accelerator, reached through the driver island, or remote inference over the network.
2. **Tool access:** an open protocol for connecting agents to tools. The Model Context Protocol has been governed by the Agentic AI Foundation under the Linux Foundation

since December 2025 (Anthropic 2025).

3. **Capability broker:** the only component that hands out authority. It grants capabilities that are attenuated, time-limited and revocable. It, not the agent, words every request a human sees. It runs isolated, yet it belongs to the trusted base: its decisions are the guarantee. It is *tested* first; the target is *proven* (Section 7).
4. **Agents:** untrusted tenants, each in its own protection domain, with a CPU budget enforced by the kernel's scheduling contexts. No guarantee depends on them.

Next to the agents, and above them in criticality, sit the **real-time loops**: control, safety monitoring and actuation. Section 4 explains why they, not the agents, have the last word.

A guarantee depends on every framed part: firmware and hardware, bootloader, kernel, static system, broker and safety controller. Agents, model access and tool access stay outside that base. Two paths cross the figure, and there is no third: authority runs from the agent through the broker, motion runs through the safety controller.

4 Real time throughout

Agents are slow and fallible. Control loops are fast and must not fail. zOS builds on that asymmetry instead of hiding it.

Time is a protected resource. In seL4 MCS, CPU time is handed out through capabilities, with budgets the kernel enforces (Heiser 2026). An agent that loops, stalls or is hijacked exhausts its own budget, not the budget of the arm controller. Memory isolation keeps a compromised agent from writing into the loop; time isolation keeps it from starving it. A CPU budget is not yet an end-to-end guarantee: memory contention, interrupts, devices and communication also decide whether a loop meets its deadline, and each needs its own bound.

The agent proposes, the verified loop disposes. Robotics described the structural pattern twenty-five years ago. In the Simplex architecture, a complex high-performance controller runs the system as long as the plant stays inside a safe envelope; a simple, high-assurance controller takes over before it leaves the region from which it can still recover (Sha 2001). zOS maps this onto its layers. The agent and its action model play the complex controller: powerful, unproven, isolated, lower in criticality. The safety controller and the switching logic play the simple one: small, at the highest criticality, and candidates for proof. The agent may suggest any motion. Only the loop moves the motor. Regulation points the same way: from January 2027, EU machinery law lists safety components whose behaviour evolves through machine learning among the machinery that needs third-party assessment (EU 2023). A safety function that stays deterministic stays out of that class.

Three goals, not one. Security: the agent cannot exceed its authority. Safety: the system violates no safety condition under stated assumptions. Usefulness: the task still gets done. None follows from the others; a controller that stops at every doubt is safe and useless. What it can reliably detect, for one robot and one task, is most of the engineering.

It has worked under attack. In DARPA's HACMS programme, the mission computer of Boeing's Unmanned Little Bird helicopter was rebuilt on seL4. In flight tests in 2017 the helicopter came under cyberattack mid-flight; compromised subsystems could not affect safe operation, and a red team given root access to the camera's virtual machine stayed confined to it (Klein et al. 2018).

Where real time is not yet provable. A real-time guarantee needs a bound on the kernel's worst-case execution time. seL4 has had a sound and complete WCET analysis, but only for the ARM11 processor. The work is archived because Arm stopped publishing worst-case instruction latencies for its out-of-order cores; open RISC-V cores such as CVA6 are named as the way forward (Trustworthy Systems n.d.-b). That analysis is now being rebuilt for the current kernel on 64-bit RISC-V, so far with placeholder instruction latencies and loop bounds (Sillitoe et al. 2026). Protection against timing side channels points the

same way: it needs a hardware instruction that so far exists only as prototypes on open RISC-V cores (Wistoff et al. 2022; Wistoff et al. 2024). Provable schedulability for MCS remains ongoing research (Heiser 2026).

The consequence is concrete. The first zOS should run on an open RISC-V core, because only there is a proof of its timing within reach.

5 Nothing to invent at the base

The base of zOS exists today as open-source work. Table 1 lists it with its state in 2026. The list mixes running software, hardware designs, protocols, tools and patterns, and the difference matters: a proof tool is not a proven broker, and a safety pattern is not a safety controller for a given robot.

The base exists; its safe integration is a substantial engineering and research task, and it carries the contribution of zOS. The **broker** must bind each permitted action to recipients, data and parameters, enforce that at every tool call, map kernel capabilities onto external credentials, define revocation for actions under way, and word approval requests in a form the agent cannot imitate, which needs a trusted path to a screen and an input device that does not run through the driver island. The **tool adapters** must leave no path around it, including the context sent to remote models. The **safety path** must be specified for a concrete robot. All three need evidence that their assumptions match the parts beneath.

Others are building this. LionsOS, from the group behind seL4, already composes seL4, Microkit and user-mode drivers into a static, modular system for cyber-physical use; a first driver, for Ethernet, has been verified with automated proofs, and the others are under way (Trustworthy Systems 2026; Tyler et al. 2026). zOS should start from it, not beside it. Agent operating systems on seL4 are appearing too: agentOS aims to give agents capabilities for tools, models and memory and to let them replace system services at run time, though its booted image does not include these services yet (Hubbard 2026); Cohesix coordinates edge AI fleets (Bower 2026); Google's KataOS paired seL4 with Rust for machine-learning devices (Google 2022). Above the kernel, designs for the broker's job are appearing: AgenticOS lets agents declare intents instead of requesting resources and derives a least-capability environment from them, so far as an architecture without an implementation (Zhao et al. 2026); Agent libOS keeps an agent's authority fixed while its tools and skills grow, on top of a conventional host (Zhang 2026). None of them names real-time control as its goal, and services that agents replace at run time, scheduling among them, rule out what a control loop needs. zOS adds the authority layer from intent to actuator, on a static, real-time base.

zOS builds entirely on open source, and we cannot say what rules open source will follow by 2031: licences written for human authors meet machine-written code, and maintainers face a flood of machine-found flaws. The licence will matter, as it did for MINIX inside Intel's Management Engine (Tanenbaum 2017). Who maintains the integration and its proofs over years matters at least as much.

6 What zOS is not, and where it can fail

An earlier draft of this design was attacked by seven AI-assisted red-team reviews, each told to tear apart one aspect: formal methods, hardware, exploitation, agent security, supply chain, systems and facts. Their findings are published with this paper. What follows is what survived as limitation.

zOS does not solve prompt injection. Capabilities limit how far a deceived agent can reach; they do not stop the deception. An injected agent can simply ask for more authority, and humans are poor gatekeepers: in one study only about 17 per cent of Android users paid attention to permissions at install time, and only 3 per cent of survey respondents answered all three comprehension questions correctly (Felt et al. 2012). Authority is also not information flow. Data can leak through channels an agent is allowed to use, which is why capability systems for agents add information-flow control (Costa et al. 2025).

Component	Existing work	State in 2026
Root of trust	OpenTitan	In production Chromebooks since March 2026 (Google 2026)
Capability hardware	CHERI, CHERIoT	Standardisation for RISC-V under way (Moore 2026); CHERIoT pairs CHERI with a clean-slate embedded OS (Amar et al. 2023)
Microkernel	seL4 with MCS	Functional correctness proven (Klein et al. 2009); MCS verified at C level on RISC-V, AArch64 targeted for Q3 2027 (seL4 Foundation n.d.-b)
System composition	seL4 Microkit	Available for use, under active development (seL4 Microkit n.d.)
Drivers	seL4 Device Driver Framework, LionsOS	Modular OS on seL4 that outperforms Linux on context-switch-intensive loads and in networking; legacy drivers in VMs (Heiser et al. 2025)
Verified components	Verus for Rust	Verifies systems code 3 to 61 times faster than prior tools (Lattuada et al. 2024)
Proof automation	AutoVerus; neuro-symbolic proof search	Over 90 % of 150 Rust proof tasks (Yang et al. 2025); 77.6 % of seL4 theorems in FVEL (He et al. 2026)
Real-time safety pattern	Simplex architecture	Established since 2001 (Sha 2001)
Robot middleware	ROS 2 on seL4	Proofs of concept for Army ground vehicles (Lide 2026)
Agent–tool protocol	Model Context Protocol	Open, under the Linux Foundation since December 2025 (Anthropic 2025)
Agent authority	CaMeL, FIDES	Capability and information-flow enforcement for agents; CaMeL solves 77 % of tasks with provable security versus 84 % undefended (Debenedetti et al. 2025; Costa et al. 2025)

Table 1. Nothing to invent: the open-source parts of zOS and their state in 2026.

zOS shrinks the blast radius. It does not make agents trustworthy.

The trusted class is larger than one would like. Firmware below the kernel, from management engines to baseboard controllers, has full access to memory and cannot be confined by any kernel. Devices that write to memory directly have defeated IOMMU protection in Linux, macOS and FreeBSD (Markettos et al. 2019). And GPUs, the hardware agents depend on most, have leaked memory between processes: one flaw exposed about 181 MB per query of a running language model (Trail of Bits 2024). zOS lists these parts as trusted and demands a hardware profile. It cannot prove them.

A proof is only as good as its seams and its specification. In three formally verified distributed systems, 11 of 16 bugs found lay in the layer between verified and unverified code, two in the specifications (Fonseca et al. 2017). zOS, built from many isolated parts, consists largely of such seams. Its consequence is a rule: the two critical paths get as few seams as possible, and each seam on them is specified as an interface that is checked when the system image is built. Verification of seL4's multikernel configuration, one kernel instance per core, is scheduled for the third quarter of 2028 (seL4 Foundation n.d.-b), so a proven zOS starts on a single core per kernel instance.

Hardening may win instead. Writing new code in a memory-safe language, without rewriting the old, works: in Android, memory-safety flaws fell below 20 per cent of all vulnerabilities in 2025 for the first time (Vander Stoep 2025). That is exactly how Linux won in 1991: good enough, incremental, invisible to users. Linux restricts processes with Landlock (Linux kernel n.d.-a) and supports real time with PREEMPT_RT (Linux kernel n.d.-b);

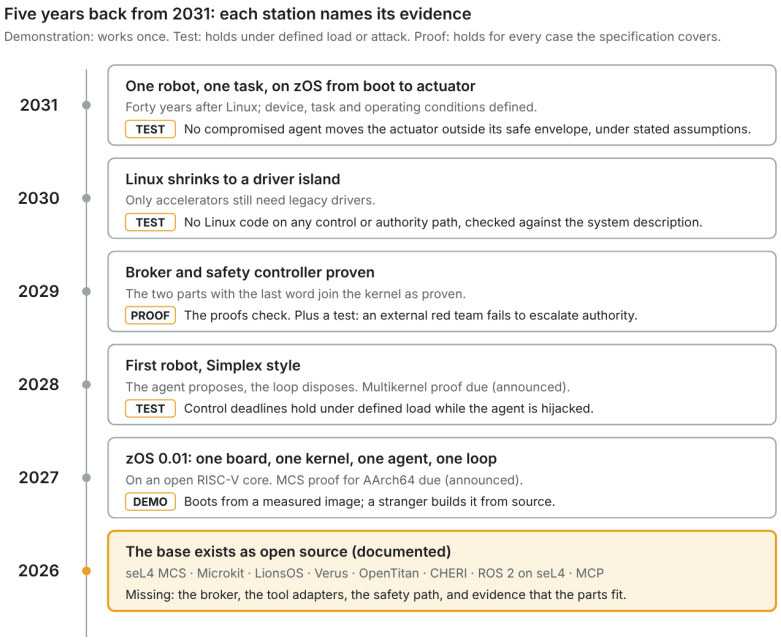


Figure 2. The path from 2026 to 2031. Each station names its evidence and whether it is a demonstration, a test or a proof; a failed check stops the path.

the ELISA project works on making Linux usable in safety applications (ELISA 2026). The obvious alternative to zOS is an agent on hardened Linux plus a separate safety controller for the motors. For the actuators that split is sound, and zOS adopts it. It leaves out the authority the agent holds over data, credentials and tools, and what it sends out, including the context shipped to remote models. A safety controller guards motion, not meaning. zOS only makes sense where hardening cannot reach: explicit authority for agents, provable timing for machines that move, and a small trusted base for systems that must not fail.

The Hurd trap. GNU Hurd, a multi-server system on Mach, never reached broad use. Its history has more than one cause, but one lesson holds for zOS: start smaller than the vision, with one board, one kernel, one agent, one loop.

7 Five years back from 2031

The path is read backwards from a deliberately modest goal: by 2031, forty years after Linux, one kind of robot doing one defined task runs on zOS from boot to actuator. Each station below is speculative, except the starting point and the verification dates the seL4 Foundation has announced. Each names its evidence and its kind: a **demonstration** shows that something works once, a **test** that it holds under defined load or attack, a **proof** that it holds for every case its specification covers. A failed red team is evidence, not proof. A station that fails its check stops the path, which makes the plan falsifiable rather than a wish.

Read from the bottom up, each station makes the next one possible. The later stations depend on dates the seL4 Foundation has announced; if those slip, the path slips. zOS 0.01 does not wait for them: on RISC-V, with one core, the MCS proof it needs exists today. The decisive station is the first: zOS 0.01 is as small as Linux 0.01 was, and just as easy to dismiss.

A possible first robot. pib, an open-source humanoid from Nuremberg with an education kit for schools, shows what this could look like; nothing of it is agreed or built. pib runs on a Raspberry Pi. Its servos are driven by small controllers with their own microcontrollers and open firmware, and a language-model agent can move its joints through MCP tools, which check the configured joint limits

in software on the Linux side (pib.rocks 2026; Tinkerforge n.d.). A path would start without new hardware: limits, a speed bound and a heartbeat timeout move into the controller firmware, the loop's last word in miniature. That firmware then holds the last word and belongs to the trusted base as *assumed*. Next comes a broker in front of the agent's tools. The Raspberry Pi 4 can carry the step after that, seL4 with pib's existing Linux as a driver island beside a native safety component (seL4 Microkit n.d.), but only as a demonstrator before zOS 0.01, for three reasons that follow from Sections 3 and 4: its kernel configuration is verified without MCS, whose proof for 64-bit Arm is not expected before 2027; its Arm cores have no published instruction latencies, so no WCET bound is possible; and its public documentation describes no IOMMU, so the Linux island, which drives devices that write to memory directly, joins the trusted base. zOS 0.01 itself runs on an open RISC-V board.

Epilogue

In 1991, Linus Torvalds did not announce an ideal system. He announced a useful one, small enough to start, open enough to grow. zOS asks for the same modesty in a different world. Its task is not to make a free Unix for a cheap PC. Its task is to make the smallest system on which an agent can act, and move, without being trusted.

None of its base is new. Capabilities date from the 1960s, the verified kernel from the 2000s, the Simplex pattern from 2001. What is new is that the reasons they stayed in niches are falling at the same time, and that the machines which need them are about to leave the screen and enter the room.

The announcement of 1991 promised a system that "won't be big". zOS makes the same promise. This time on purpose.

Glossary

Trust and proof

Byzantine component. A part that may fail in any way at all, including by lying or acting against the system on purpose. *In zOS:* the assumption made about every agent (Section 1).

Formal verification. Proving with mathematics, checked by a computer, that a program does what its specification says, for every possible input. *In zOS:* the standard for the parts a guarantee depends on; seL4's kernel already meets it (Sections 2.3, 3).

Functional correctness. The proven statement that an implementation behaves exactly as its specification describes. *In zOS:* what seL4 has proven for its verified configurations (Section 3).

Proven, tested, assumed. Three levels of knowledge about a part: shown for every case its specification covers, shown to hold under defined load or attack, or simply trusted. *In zOS:* every part of the trusted base carries one of these labels (Section 1).

Reference monitor. The mechanism that checks every access; it must always be invoked, be tamper-proof and be small enough to analyse and test. *In zOS:* the reason the system must stay small (Section 1).

Seam. The place where a proven part meets an unproven one, and where the assumptions of one must match the guarantees of the other. *In zOS:* kept to a minimum on the two critical paths, and each one is specified (Section 6).

Trusted base. The parts that must work correctly for a guarantee to hold; if one of them fails, the guarantee fails with it. *In zOS:* kept small and listed part by part (Sections 1, 3).

Kernel and time

Microkernel. A kernel that does only what must run with full privilege, such as isolation, messages and scheduling, and leaves everything else to ordinary programs. *In zOS:* seL4 (Section 3).

seL4. A microkernel from the L4 family and the first general-purpose kernel with a machine-checked proof of functional correctness. *In zOS:* the kernel (Sections 2.1, 3).

Mixed criticality (MCS). Running parts that matter to different degrees on one ma-

chine, so that a less important part can never take time from a more important one. *In zOS:* seL4's MCS extensions make processor time a protected resource (Sections 3, 4).

Scheduling context. In seL4 MCS, a kernel object that grants a program a budget of processor time per period; the kernel enforces it. *In zOS:* limits how much time an agent can use (Sections 3, 4).

Real time. Computing in which a result is only correct if it arrives before its deadline. *In zOS:* required for every loop that moves something; provable timing is a goal, not yet a property (Sections 1, 4).

WCET. Worst-case execution time: a safe upper bound on how long a piece of code can take, needed to prove that deadlines hold. *In zOS:* exists for seL4 only on an old Arm core; being rebuilt for RISC-V (Section 4).

Timing channel. A leak in which one program learns secrets from how long another program's operations take. *In zOS:* closed only with hardware support that exists as prototypes on open RISC-V cores (Section 4).

Protection domain. An isolated program in a static seL4 system, with its own memory and only the access it was given. *In zOS:* every agent and every component runs in its own (Section 3).

Static system. A system whose structure, programs, channels and shared memory are fixed when the image is built and cannot change at run time. *In zOS:* the whole system is declared in one description (Section 3).

Microkit. seL4's framework for building static systems from protection domains, channels and shared memory. *In zOS:* turns the system description into the bootable image (Section 3).

LionsOS. A modular, static operating system on seL4 and Microkit from the group behind seL4. *In zOS:* the base to start from (Section 5).

Multikernel. A configuration that runs one separate kernel instance per processor core instead of one kernel for all cores. *In zOS:* how a proven system will later use several cores (Section 6).

Agents and authority

Agent. A program that reads, decides and acts on its own, usually driven by a language model. *In zOS:* an untrusted tenant with a time budget (Sections 1, 3).

Capability. An unforgeable token that both names a resource and grants a specific right to it; whoever does not hold it cannot use the resource. *In zOS:* the only way authority is passed (Sections 2.4, 3).

Capability broker. The single component that hands out authority: limited, time-bound and revocable. *In zOS:* part of the trusted base; tested first, proven later (Sections 3, 5).

Confused deputy. A program that misuses authority given to it by one party while acting on behalf of another. *In zOS:* the old name for what prompt injection does to agents (Section 2.4).

Prompt injection. Instructions hidden in data, such as a web page or an email, that steer an agent reading it. *In zOS:* not solved, but limited in what it can reach (Sections 2.4, 6).

Information-flow control. Tracking where data comes from and where it may go, so that it cannot leave through a channel that is otherwise allowed. *In zOS:* needed in addition to capabilities (Section 6).

Model Context Protocol (MCP). An open protocol that connects agents to tools and data sources. *In zOS:* the way agents reach tools, always through the broker (Section 3).

Robots and safety

Action model (VLA). A model that turns what a robot sees and is told into motor commands; VLA stands for vision-language-action. *In zOS:* plays the complex, unproven controller (Sections 2.4, 4).

Action chunking. Emitting several motor commands at once, so that the robot keeps moving while the model computes the next ones. *In zOS:* why the motion path needs real time (Section 2.4).

Simplex architecture. A safety pattern in which a complex controller runs the system

while a simple, trustworthy one takes over before the system leaves a safe region. *In zOS:* the agent proposes, the verified loop disposes (Section 4).

Safety controller. The simple controller in the Simplex pattern, with the last word over every motion. *In zOS:* part of the trusted base, at the highest criticality (Sections 3, 4).

ROS 2. The Robot Operating System, the most common middleware that connects the software parts of a robot. *In zOS:* shown running on seL4 in proofs of concept (Section 2.4).

Hardware

RISC-V and CVA6. RISC-V is an open instruction set anyone may implement; CVA6 is an open-source RISC-V core whose timing can be fully inspected. *In zOS:* the first platform, because only there is a timing proof within reach (Sections 4, 7).

Root of trust. A small piece of hardware that is trusted to measure and check the first code a machine runs. *In zOS:* the bottom of the trusted base; OpenTitan is an open one (Section 3).

OpenTitan. An open-source silicon root of trust, in production since 2026. *In zOS:* the open option for the root of trust (Sections 3, 5).

IOMMU. Hardware that restricts which memory a device may read and write directly (DMA, direct memory access). *In zOS:* without a proven IOMMU configuration, a driver with such a device joins the trusted base (Sections 3, 7).

Firmware. Code that runs below or before the operating system, often with more privilege than the kernel. *In zOS:* trusted and assumed; it cannot be confined by any kernel (Sections 3, 6).

CHERI. A hardware extension that makes pointers unforgeable, so that memory errors are caught by the processor. *In zOS:* used where available (Sections 3, 5).

Driver island. A Linux virtual machine that runs drivers zOS cannot replace, such as those for graphics and AI accelerators. *In zOS:* on the agent's side, outside the trusted base where an IOMMU confines its devices (Section 3).

References

Amar, S. et al. (2023): CHERIoT: Complete Memory Safety for Embedded Devices. Proc. MICRO '23, 641–653.

Anderson, J. P. (1972): Computer Security Technology Planning Study. Vol. II, ESD-TR-73-51, U.S. Air Force Electronic Systems Division.

Anthropic (2025): Donating the Model Context Protocol and establishing the Agentic AI Foundation. 9 December 2025.

Anthropic (2026a): Project Glasswing: Securing critical software for the AI era. 7 April 2026.

Anthropic (2026b): Project Glasswing: An initial update. 22 May 2026.

Apple (n.d.): XNU kernel source, README. Accessed 19 October 2026.

Black, K.; Galliker, M. Y.; Levine, S. (2025): Real-Time Execution of Action Chunking Flow Policies. arXiv:2506.07339.

Bower, L. (2026): Cohesix: a research operating system for edge AI. Project page, accessed 20 October 2026.

Costa, M. et al. (2025): Securing AI Agents with Information-Flow Control. arXiv:2505.23643.

DARPA (2025): AI Cyber Challenge marks pivotal inflection point for cyber defense. August 2025.

Debenedetti, E. et al. (2025): Defeating Prompt Injections by Design. arXiv:2503.18813.

Dennis, J. B.; Van Horn, E. C. (1966): Programming Semantics for Multiprogrammed Computations. Communications of the ACM 9(3).

DiBona, C.; Ockman, S.; Stone, M. (eds.) (1999): Open Sources: Voices from the Open Source Revolution, Appendix A: The Tanenbaum–Torvalds Debate. O'Reilly.

ELISA (2026): Enabling Linux in Safety Applications project expands premier membership with NVIDIA. Linux Foundation, 26 February 2026.

EU (2023): Regulation (EU) 2023/1230 of the European Parliament and of the Council on machinery. Official Journal of the European Union; applies from 20 January 2027; Annex I, Part A, point 5.

Felt, A. P.; Ha, E.; Egelman, S.; Haney, A.; Chin, E.; Wagner, D. (2012): Android Permissions: User Attention, Comprehension, and Behavior. UC Berkeley, UCB/EECS-2012-26.

Fonseca, P. et al. (2017): An Empirical Study on the Correctness of Formally Verified Distributed Systems. Proc. EuroSys '17.

Google (2022): Announcing KataOS and Sparrow. Google Open Source Blog, October 2022.

Google (2025): Building the next generation of physical agents with Gemini Robotics-ER 1.5. 25 September 2025.

Google (2026): OpenTitan shipping in production. Google Open Source Blog, March 2026.

Greshake, K. et al. (2023): Not what you've signed up for: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection. Proc. AISec '23, 79–90. arXiv:2302.12173.

Hardy, N. (1988): The Confused Deputy (or why capabilities might have been invented). ACM SIGOPS Operating Systems Review 22(4), 36–38.

He, B.; Li, Z.; Sun, W.; Yao, Y.; Chen, T.; Ma, X.; Su, Z. (2026): Neuro-Symbolic Proof Generation for Scaling Systems Software Verification. Proc. OSDI '26. Preprint arXiv:2603.19715.

Heiser, G. (2026): MCS safety – an OS perspective. Award talk, RTAS '26.

Heiser, G.; Elphinstone, K. (2016): L4 Microkernels: The Lessons from 20 Years of Research and Deployment. ACM Transactions on Computer Systems 34(1).

Heiser, G. et al. (2025): Fast, Secure, Adaptable: LionsOS Design, Implementation and Performance. arXiv:2501.06234.

Hubbard, J. (2026): agentOS: an operating system designed from the core up for agents. Source repository, accessed 20 October 2026.

Klein, G. et al. (2009): seL4: Formal Verification of an OS Kernel. Proc. SOSP '09.

Klein, G.; Andronick, J.; Kuz, I.; Murray, T.; Heiser, G.; Fernandez, M. (2018): Formally Verified Software in the Real World. Communications of the ACM 61(10), 68–77.

Lamport, L.; Shostak, R.; Pease, M. (1982): The Byzantine Generals Problem. ACM Transactions on Programming Languages and Systems 4(3), 382–401.

Lattuada, A. et al. (2024): Verus: A Practical Foundation for Systems Verification. Proc. SOSP '24.

Leemhuis, T. (2025): Zahlen, bitte! Linux – soon to weigh 40 million lines of code. heise online, January 2025.

Lide, D. (2026): ARSENAL: Assured Robotics Embedded Systems and Applications in Ground Vehicles. Talk, seL4 Summit 2026, Vancouver.

Liedtke, J. (1993): Improving IPC by Kernel Design. Proc. SOSP '93, 175–188.

Linux kernel (n.d.-a): Landlock: unprivileged access control. Kernel documentation, accessed 20 October 2026.

Linux kernel (n.d.-b): Real-time preemption (PREEMPT_RT). Kernel documentation, accessed 20 October 2026.

Markettos, A. T. et al. (2019): Thunderclap: Exploring Vulnerabilities in Operating System IOMMU Protection via DMA from Untrustworthy Peripherals. Proc. NDSS 2019.

Miller, M. S. (2006): Robust Composition: Towards a Unified Approach to Access Control and Concurrency Control. PhD thesis, Johns Hopkins University.

Moore, S. W. (2026): CHERI is being standardised, so why do we still need to do research? CHERI Blossoms 2026, Cambridge, 27 March 2026.

Nasr, M. et al. (2026): The Attacker Moves Second: Stronger Adaptive Attacks Bypass Defenses Against LLM Jailbreaks and Prompt Injections. Proc. USENIX Security '26.

Parker, L. (2022): The seL4 Device Driver Framework. Talk, 4th seL4 Summit, Munich.

pib.rocks (2026): pib-backend: software for the pib robot. Source repository, accessed 20 October 2026.

seL4 Foundation (n.d.-a): Verified Configurations. Accessed 19 October 2026.

seL4 Foundation (n.d.-b): seL4 Roadmap. Accessed 19 October 2026.

seL4 Microkit (n.d.): Microkit User Manual, v2.3.0. Accessed 19 October 2026.

Sha, L. (2001): Using Simplicity to Control Complexity. IEEE Software 18(4), 20–28. doi:10.1109/MS.2001.936213.

Sillitoe, S.; Basit, A.; Giacometti, M.; Heiser, G. (2026): Resurrecting seL4's WCET Analysis. OSPERT 2026.

Tanenbaum, A. S. (2017): An Open Letter to Intel. Vrije Universiteit Amsterdam.

Tinkerforge (n.d.): Servo Bricklet 2.0 firmware. Source repository, accessed 20 October 2026.

Torvalds, L. (1991): Posting to comp.os.minix, 25 August 1991. Reproduced in iTWire.

Trail of Bits (2024): LeftoverLocals: Listening to LLM responses through leaked GPU local memory. 16 January 2024.

Trustworthy Systems (2026): LionsOS. Project page, accessed 20 October 2026.

Trustworthy Systems (n.d.-a): Proof Statistics. UNSW Sydney.

Trustworthy Systems (n.d.-b): WCET: Worst-Case Execution-Time Analysis of seL4. UNSW Sydney.

Tyler, S.; Legnani, A.; Zhao, J.; Shen, Z.; Wu, R.; Tanaka, M.; Heiser, G. (2026): SMT-based Deductive Verification of Device Drivers using the Pancake-to-Viper Transpiler. Proc. PLOS '26, Prague.

Vander Stoep, J. (2025): Rust in Android: move fast and fix things. Google, 13 November 2025.

W3C (2016): 25 years ago the world changed forever. W3C Blog, August 2016.

Wistoff, N.; Schneider, M.; Gürkaynak, F. K.; Heiser, G.; Benini, L. (2022): Systematic Prevention of On-Core Timing Channels by Full Temporal Partitioning. doi:10.1109/TC.2022.3212636.

Wistoff, N.; Heiser, G.; Benini, L. (2024): fence.t.s: Closing Timing Channels in High-Performance Out-of-Order Cores through ISA-Supported Temporal Partitioning. APPLEPIES 2024, Turin.

Yang, C. et al. (2025): AutoVerus: Automated Proof Generation for Rust Code. Proc. OOPSLA '25.

Zhang, Y. (2026): Agent libOS: A Runtime Substrate for Capability-Controlled Self-Evolving LLM Agents. arXiv:2606.03895.

Zhao, Z. et al. (2026): AgenticOS: An Intent-Oriented Secure Operating System Architecture for Autonomous AI Agents. arXiv:2606.21129.